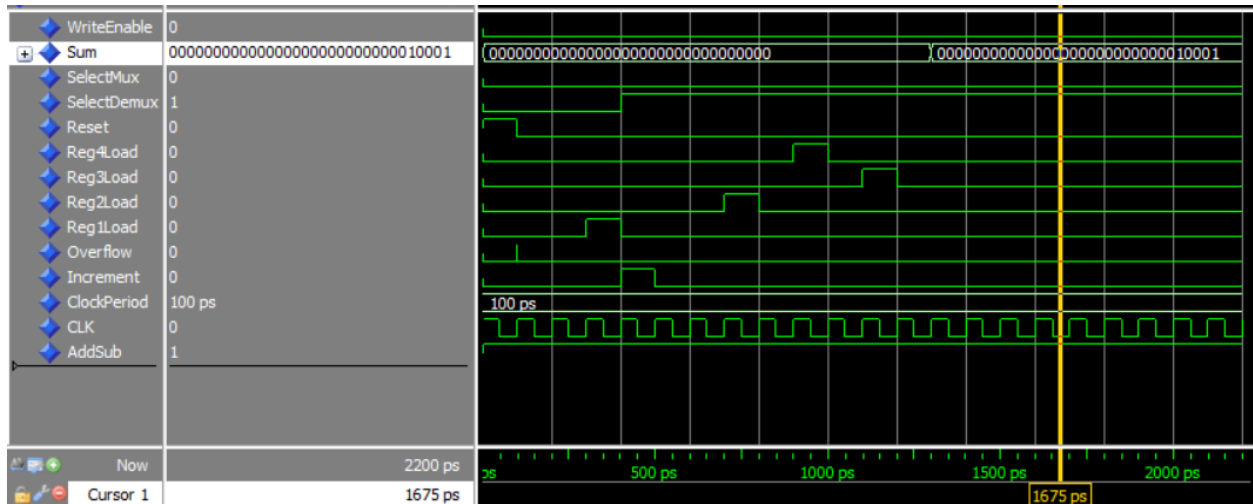
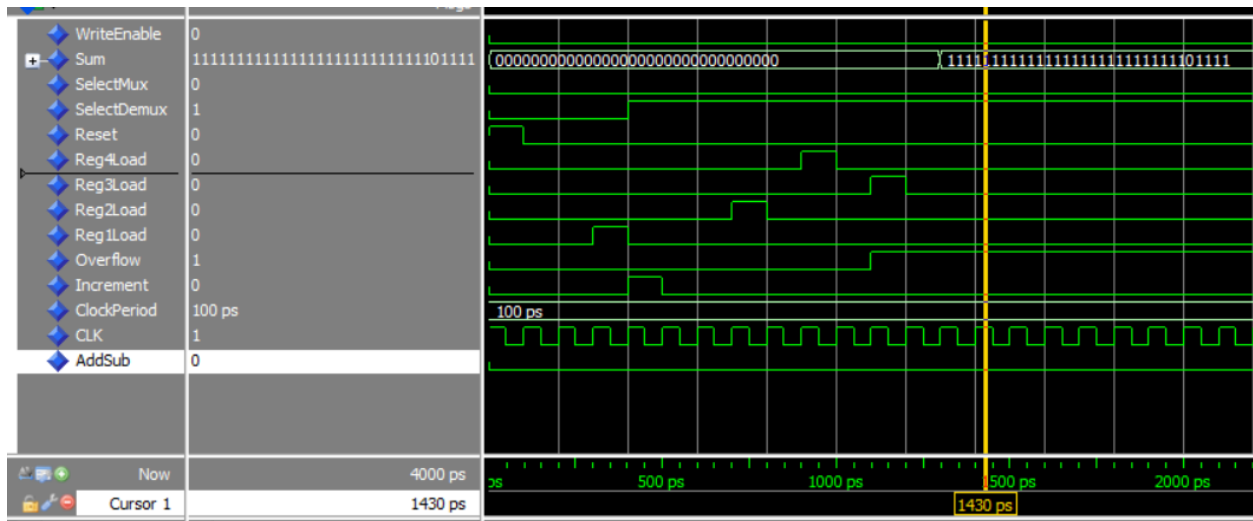


Allan Yunayev Final Project

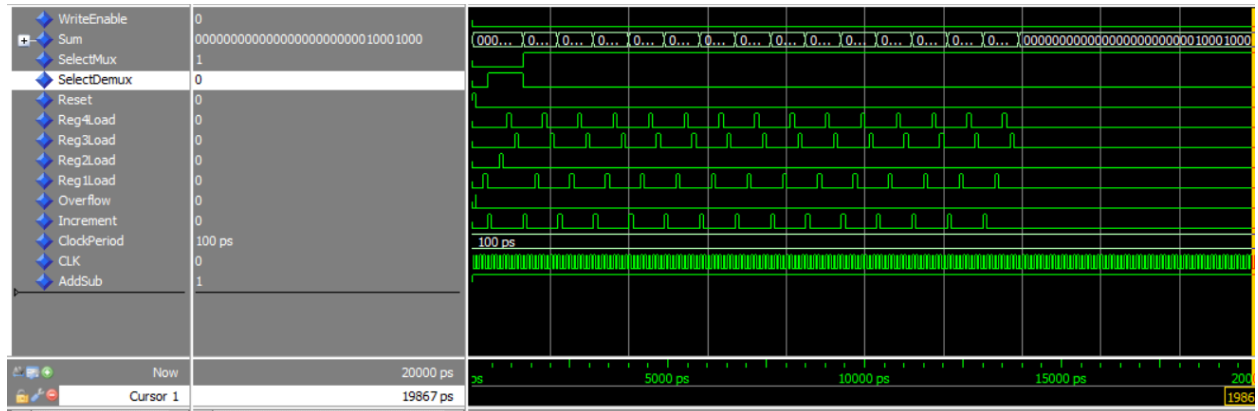
Test Bench Results for Two Sum 000000000000000000000000000000 and
00000000000000000000000000000010001 = 00000000000000000000000000000010001;



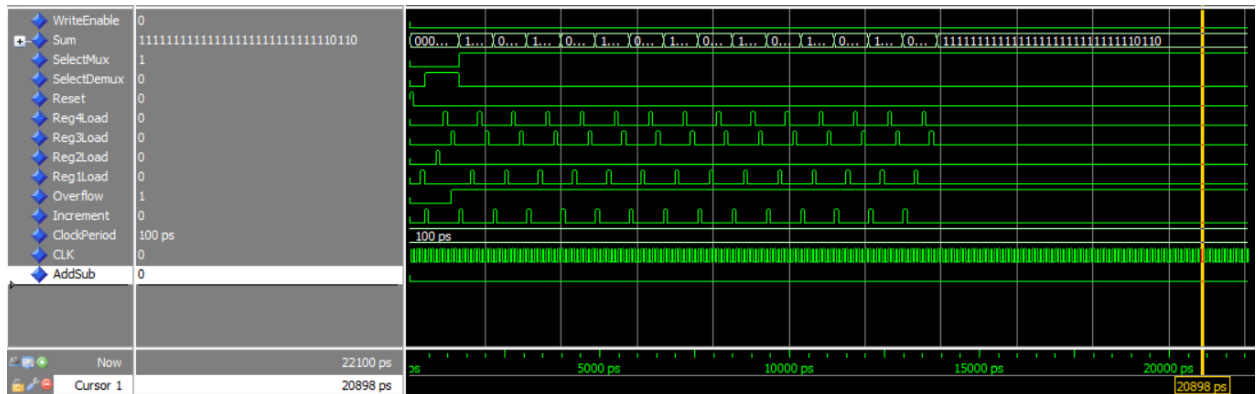
Test Bench Results for Two Sub 000000000000000000000000000000 –
00000000000000000000000000000010001 = 11111111111111111111111111111101111



Test Bench of 10 ADD $0+17+2+3+4+5+6+7+8+9+10+11+12+13+14+15=136 \Leftrightarrow 10001000$ I
 accidentally did all 16 just realized that too lazy to change that though



TestBench for 10 Sub $(15-(14-(13-(12-(11-(10-(9-(8-(7-(6-(5-(4-(3-(2-(0-17))))))))))))))=-10$
 $\Leftrightarrow 111111111111111111111111111110110$
 accidentally did all 16 just realized that too lazy to change that though



The screenshot displays the Logic Analyzer interface. On the left, a list of signals is shown, each with a blue diamond icon indicating it is active. The signals are: WriteEnable, Sum, SelectMux, SelectDemux, Reset, Reg4Load, Reg3Load, Reg2Load, Reg1Load, Overflow, Increment, ClockPeriod, CLK, AddSub, and Ram[q_a]. The right side of the image shows a timing diagram for these signals. The horizontal axis represents time in picoseconds (ps), with major ticks every 100 ps from 3900 ps to 14600 ps. A yellow vertical cursor is positioned at 14553 ps. The signals are represented by green and black waveforms on a grid. The 'Sum' signal is a constant high (black) line. The 'SelectMux' signal is a constant low (green) line. The 'SelectDemux' signal is a constant high (black) line. The 'Reset' signal is a constant low (green) line. The 'Reg4Load' signal is a constant low (green) line. The 'Reg3Load' signal is a constant low (green) line. The 'Reg2Load' signal is a constant low (green) line. The 'Reg1Load' signal is a constant low (green) line. The 'Overflow' signal is a constant low (green) line. The 'Increment' signal is a constant low (green) line. The 'ClockPeriod' signal is a constant low (green) line. The 'CLK' signal is a constant low (green) line. The 'AddSub' signal is a constant low (green) line. The 'Ram[q_a]' signal is a constant low (green) line.

[illegible]